

DATA MINING & MACHINE LEARNING

Predicting Online Shoppers' purchasing Intention

Data Mining & Machine Learning Project

M.Sc. in Artificial Intelligence and Data Engineering — Academic Year 2025–2026

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Presentation Outline

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Problem & Dataset

Task definition · features · EDA

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Pipeline & Experiments

Preprocessing · model selection · hyperparameter optimization

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Metrics · cross-validation · final model performance

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Error analysis · limitations · future directions

PART 1

PART 2

PART 1 — TECHNICAL

Problem, Pipeline & Results

Problem definition · feature engineering · model selection
hyperparameter optimization · evaluation · test-set results

Problem Definition & Dataset

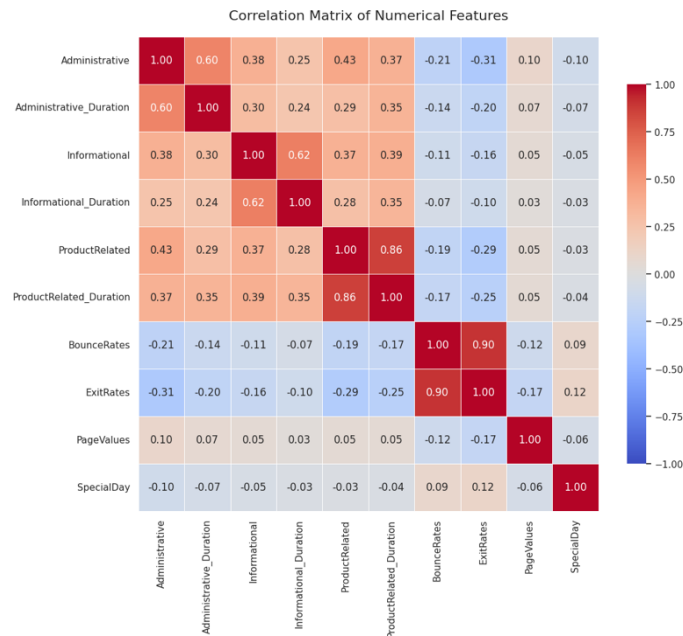
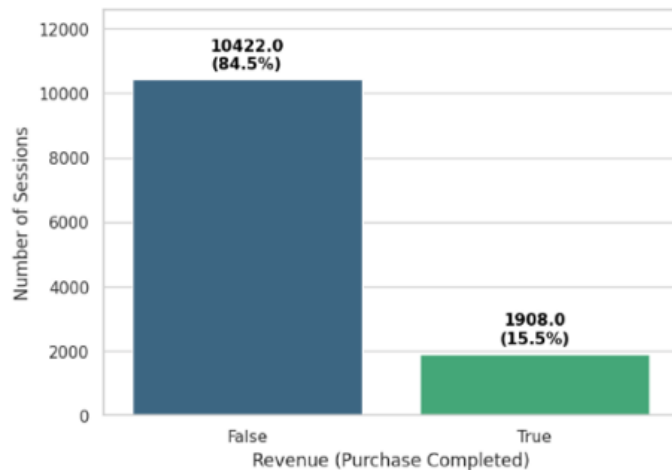
The Problem

- Predict whether a browsing session ends in a purchase, using only web-analytics session data.
- Task type: binary classification of the Revenue target (purchase / no purchase).
- Practical value: spotting likely buyers lets e-commerce target discounts and assistance to secure the sale.
- Key challenge: strong class imbalance → only ~15% of sessions convert and missing a real buyer costs more than a false alarm.

Dataset at a Glance

Task	Binary classification
Dataset	UCI Online Shoppers - 12,330 samples, 17 features
Target	Revenue — session ended in a transaction (True/False)
Features	10 numerical, 7 categorical
Missing values	0 missing values
Class balance	False: 84.5%, True: 15.5% - unbalanced
Split	10×10 Nested Cross-Validation
Metric(s)	F1-score and particularly recall

Exploratory Data Analysis



Key findings from EDA:

- Strong class imbalance: only 15.5% of sessions convert vs 84.5% non-purchase → this motivates SMOTENC oversampling.
- Two highly correlated feature pairs → BounceRates–ExitRates and ProductRelated–ProductRelated_Duration.
- No missing values across all 12,330 sessions, so no imputation step is required.

Feature Engineering & Selection

Engineering

- No new features engineered.
- Numerical features standardized with StandardScaler.
- One-hot encoding; OS, Browser, Region, TrafficType cast to strings.
- No imputation needed (0 missing).
- 125 duplicate sessions removed.

Selection Method

- Filter method: prune redundant features from correlated pairs.
- Criterion: Pearson r and p -value of each feature vs Revenue.
- Within each pair, keep the feature more correlated with the target.
- BounceRates and ProductRelated_Duration dropped.
- 15 features kept.

Selection Evidence

```
Analyzing Pair: [ProductRelated] vs [ProductRelated_Duration]
> ProductRelated:
  - Correlation: 0.1560
  - P-value: 2.2083e-67
> ProductRelated_Duration:
  - Correlation: 0.1501
  - P-value: 2.0686e-62

-----
Analyzing Pair: [BounceRates] vs [ExitRates]
> BounceRates:
  - Correlation: -0.1451
  - P-value: 2.0740e-58
> ExitRates:
  - Correlation: -0.2043
  - P-value: 3.5455e-115
```

Pearson r and p -value vs Revenue: ExitRates beats BounceRates; ProductRelated beats its duration.

ML Pipeline: Avoiding Data Leakage & Bias



Data Leakage Prevention

- All preprocessing fitted only on training folds
- Whole workflow wrapped in one `ImbPipeline`: `SMOTENC` → `ColumnTransformer` → classifier.
- Oversampling and scaling fitted per fold; validation/test folds never seen.

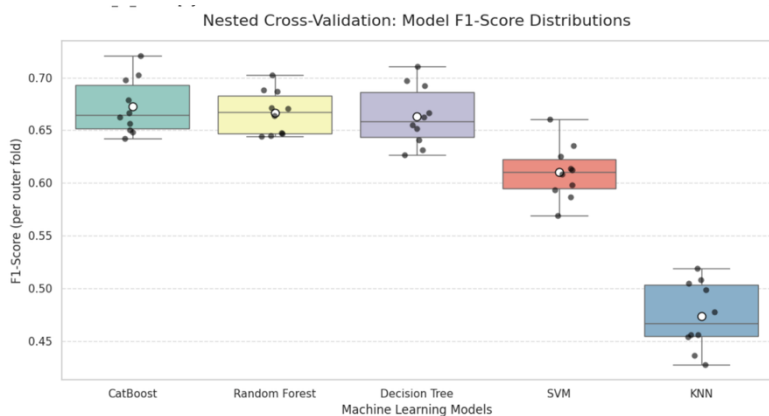
Class Imbalance Handling

- `SMOTENC` oversampling (handles mixed numerical + categorical features).
- Placed before the `ColumnTransformer`; fitted only on each training fold.
- Ablation without `SMOTENC` raised Accuracy but lowered Recall and F1.

Metric Justification

- Primary metric: F1-score → balances Precision and Recall on imbalanced data.
- A missed buyer (FN) costs more than a wasted discount (FP) → favour Recall.
- Accuracy, Precision also reported for completeness.

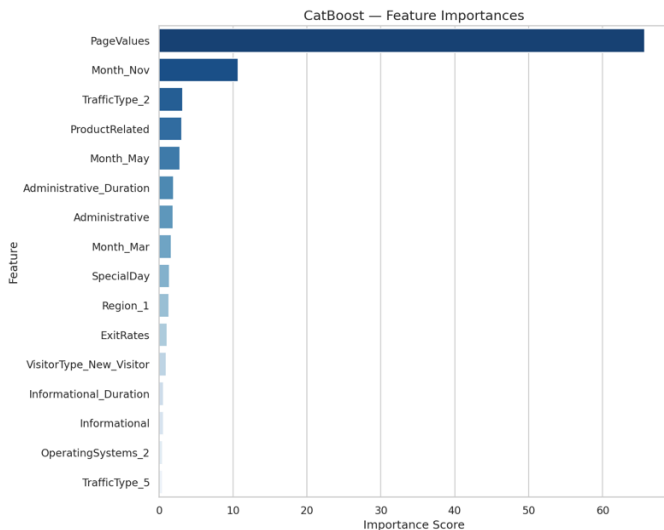
Model Selection & Hyperparameter Optimization



Key Decisions

Candidate models	Decision Tree, Random Forest, SVM, KNN, CatBoost
CV strategy	10×10 Stratified Nested CV (outer test, inner GridSearch)
Selection criterion	Mean F1-score across the 10 outer folds
Chosen model	CatBoost → highest mean F1 (0.6726) and strong Recall
Search strategy	Grid Search in the CV loop
Search space	iterations {100,200}, learning_rate {0.01,0.05}, depth {3,5}
Best config	iterations=200, learning_rate=0.01, depth=5
Statistical test	Wilcoxon Signed-Rank on the 10 outer-fold F1-Score distributions CatBoost vs Random Forest, $p=0.0488$

Feature Importance & Target Leakage



Why the GA Features Were Removed

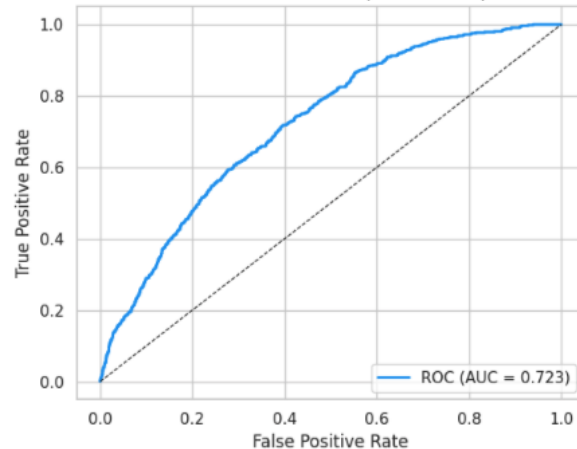
- Initial CatBoost importance is dominated by PageValues, a Google Analytics metric.
- PageValues and ExitRates come from opaque GA aggregation → risk of indirect target leakage.
- Prior work [1] exploits these metrics for high accuracy; we exclude them for unbiased features.
- Ablation removing all GA features still reaches $F1 = 0.37$ on this imbalanced task.
- XAI is therefore applied to the ablated model to reveal the true purchase drivers.

Test-Set Results

Confusion Matrix — Test Set (no-GA model)

True label	Predicted label	
	No Purchase	Purchase
No Purchase	2307	783
Purchase	252	320

ROC Curve — Test Set (no-GA model)



Key Metrics (Test Set)

0.37

F1-score

0.56

Recall

0.723

ROC-AUC

0.50

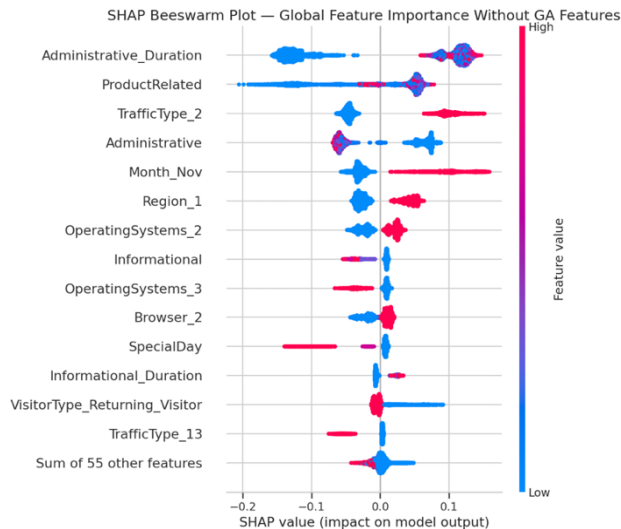
Random AUC

PART 2 — CRITICAL ANALYSIS

Explainability & Critical Discussion

How does the model work? · What did it get wrong and why?
Limitations · future directions · domain impact

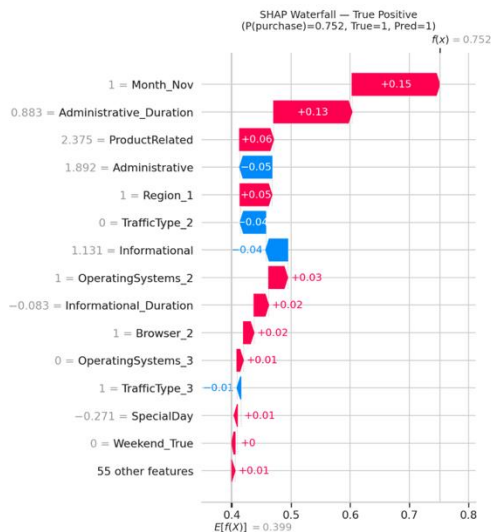
Global Explainability: What Drives the Model Overall?



Interpretation

- We use SHAP with a TreeExplainer, because CatBoost is a tree-based gradient-boosting model.
- The top drivers make sense for an online shop.
Administrative_Duration is the strongest: more time on account/checkout pages means the user is closer to buying.
Month_Nov is a strong positive seasonal driver (Black Friday).
- SpecialDay** goes the "wrong" way: sessions close to public holidays are *less* likely to convert, users browse but buy less.
- TrafficType_2** push toward purchase, consistent with the domain.

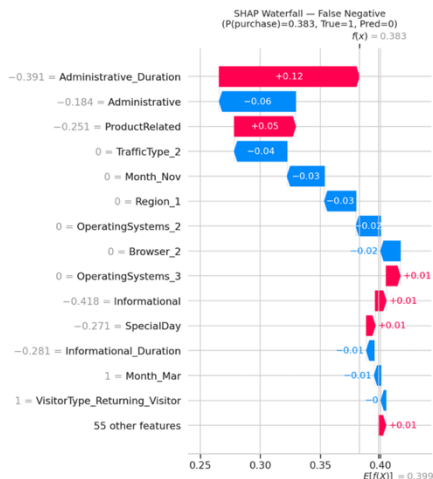
Local Explainability: Why This Specific Prediction?



Case Analysis

- The sample is a randomly selected True Positive session (purchase correctly predicted).
- Month_Nov and above-average Administrative_Duration are the strongest positive drivers.
- The two features that drive this one prediction are exactly the ones at the top of the global Beeswarm, the model uses the same logic for the individual as for the whole population
- It is consistent with the domain knowledge: a buyer reached through seasonal context and administrative engagement.

Error Analysis: What Did the Model Get Wrong and Why?



Reverse-Engineering Errors

- The shared pattern is that these are real buyers whose contextual markers are all missing at the same time, they simply do not look like the typical buyer the model learned.
- These samples show a low administrative page count, no high-converting TrafficType_2, non-November visit. So the session lacks all three of its strongest positive cues at once.
- The waterfall shows the prediction is pulled down by a cascade of small negative contributions rather than one big mistake.
- This is a genuinely hard case: the user converted despite a profile that statistically points to no purchase
- The model leans on seasonal/contextual cues, so off-season buyers are systematically harder.

Sakar et al. [1] report higher accuracy, but they exploit Google Analytics metrics (PageValues, ExitRates). We removed these to avoid indirect target leakage, which explains our lower F1 (0.37) on purely behavioural features.

Domain Impact, Limitations & Future Directions

Domain Impact

- This system could be used as a real-time conversion-assist tool for an online shop.
- It supports Real-time prioritisation of high-intent visitors during a session.
- A missed buyer (FN) costs more than a wasted incentive (FP) → favour Recall.
- A medium level of trust is required, it is a decision-support tool.

Limitations

- Single public dataset with aggregated session counts only.
- Performance leaned on opaque GA metrics; removing them sharply lowered F1.
- No event ordering, so intra-session intent dynamics are not captured.
- Evaluated on one dataset, with no real-world deployment test.

Future Directions

- Richer behavioural data such as cart and click events.
- Transparent, leakage-free engagement features.
- Sequence models (LSTM / Transformer) on raw session logs.

Thank You

Conclusions:

- Predicted online purchase intention as binary classification on the imbalanced UCI Online Shoppers dataset.
- CatBoost won a 10×10 Nested CV comparison, confirmed significant by a Wilcoxon test ($p = 0.0488$).
- After removing leaky GA features, SHAP shows November seasonality and administrative engagement drive real purchases.
- Pure behavioural signals are weaker ($F1 = 0.37$); next: richer cart/event features and sequence models.